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Fallacies

Introduction

If you were an Artificial Intelligence instead of a human being, the last couple of weeks would've given you lots of tools for identifying strong and weak arguments (albeit at an introductory level). You've learned a technique for uncovering the structure of arguments, then learned about evaluating premises and logic. So, in principle, you know how to construct and reconstruct arguments, and tell good arguments from bad ones. What you need now is a commitment to *developing* a critical mindset and practice. As you keep using the techniques taught in this course, you'll become better and better at spotting errors in arguments.

We can speed up that process by warning you of some forms of argumentative error and failures in thinking clearly play upon human psychology and therefore common enough that you can be sure you'll encounter them in yourself and others. You already have the tools to detect these weak forms of reasoning, but knowing that you can expect these particular errors will help you recognize them more quickly. In some cases, simply recognizing them will be enough to help you avoid them; in other cases, you'll need to take additional steps.

The Roots of Common Errors

Virtually everybody recognizes that human beings are fallible—that is, we're all capable of making errors. That's true in any aspect of life, including reasoning about what we should believe and how we should act. This is something we all know, but it's also something we're much more likely to be conscious of when we think about others than when we think about ourselves. Keeping our own fallibility in focus is an important part of developing a critical mindset.

But the fact we're fallible doesn't explain why certain mistakes are much more common than others. What does explain it is the fact that our psychology functions in ways that make avoiding certain mistakes particularly difficult. This shouldn't be surprising, as other aspects of our lives are similar. For example, it's not the fact that we're fallible in our ability to manipulate objects with our hands that explains why we more often fail at shooting a bull's eye in darts than we do at placing a glass of water on a coaster; rather, it's more specific facts about how our bodies work that explain why shooting a bull's eye is more difficult than safely putting down a drink.

Philosophers have long analyzed and classified faulty forms of argument. The traditional approach was to focus on the domain of logic and argumentation theory by looking at an argument and asking *What is wrong with* it*?* This is an important question, and the focus of the centuries-old study of *argumentative fallacies*. But equally important is the question *Why do people fall for these faulty arguments?* (or, if you will, *What is wrong with* us*?*). This is what the more recent research

program on *cognitive biases* tries to answer. We already discussed the notion of cognitive biases in the introduction to this book. Generally speaking, these are shortcuts in reasoning that are useful in one way or another, but that systematically lead us astray when applied in inappropriate contexts.

Cognitive biases, then, aren't the same as argumentative fallacies. Fallacies are bad arguments; cognitive biases are better thought of as aspects of our psychology that make us prone to accept bad arguments. Moreover, some cognitive biases actually have nothing to do with arguments, instead operating at a different level: they might influence our sense-perception, or whether we find an idea intuitively plausible independent of any arguments being made. At the same time, argumentative fallacies are related to cognitive biases. There's an infinite number of bad argument structures, but the number of recognized fallacies is much smaller (though still surprisingly large). Recognized fallacies are bad arguments that occur frequently. They wouldn't be so common if they weren't psychologically persuasive—something about these fallacies makes it tempting for human psychology to make or accept such arguments. All this means that the line between fallacies and biases is somewhat blurry. Sometimes fallacies and biases work hand in hand, and some of the phenomena we discuss as fallacies could arguably be classified as biases and vice versa.

With that proviso out of the way, let's turn to fallacies. We'll cover cognitive biases in a sperate text.

Argumentative Fallacies

Argumentative fallacies are faulty arguments, errors in reasoning that lead to weakly or unjustified conclusions. Among the many ways in which arguments can go wrong, some occur often enough that we've given them names. In fact, we already looked at two of them last week: denying the antecedent and affirming the consequent. The list of recognized and named fallacies has grown very long over centuries of philosophical inquiry, but we're going to limit ourselves here to just eight of the most prominent ones. You're very likely to have encountered most of these fallacious argument forms before (even if you didn't know they had names) and you'll certainly encounter them again. Knowing what exactly is wrong with them will make you less likely to be fooled by them (and knowing their names will help you remember what's wrong with them).

Fallacies That Use Unacceptable Premises

Sometimes, people talk as if fallacies are always failures of logic, but that's not quite accurate. There are some fallacies that are problematic because they start with unacceptable premises—premises that aren't worthy of being accepted, and so provide a poor foundation for an argument. The three fallacies discussed immediately below are of this type.

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Straw Man

The *straw man* fallacy occurs when an arguer holds up a distorted or weakened version of an opponent's view or argument and then "destroys" it with criticism. In other words, rather than taking the opponent's argument seriously, the arguer who uses a straw man scores easy points by taking aim at something the opponent didn't actually say. What makes this fallacy attractive is that arguing against a weakened version of an opponent's argument requires less mental effort. It also allows the arguer to make what, on a superficial level, looks like a more convincing case against their opponent, which means it also requires less effort for any listeners to take sides.

Imagine someone objecting to a new policy that would require all workers in your warehouse and in your office to wear steel-toed safety shoes. But instead of giving reasons why the new policy is wrongheaded, she distorts the aims of the policy like so: "Administration just wants to control what we wear. They want us to look like identical robots and destroy our individuality completely." If this were really the policy's aim, it could be rightly criticized as unethical. After all, freedom is important. But how likely is it that this is really the goal of the policy? It doesn't insist on complete uniformity; it merely insists on uniformity in one regard, leaving open other avenues for self-expression. But rather than thinking hard about what the policy's real objective is, and what it really requires of workers, the arguer here is trying to convince listeners by making the policy sound both unreasonable and grounded in bad intentions. Such an argument is easy, but it's a mistake.

The unacceptable premise that's characteristic of the straw man fallacy is a claim about what somebody else has said, done, or intended. This premise is typically a violation of at least one of the principles of fidelity and charity we discussed in chapter 6.

It's worth noting that there are two different kinds of straw man argument. We can refer to one type as the *personal straw man.* A personal straw man is an argument that distorts the words, intentions, or arguments of a particular person or institution. Then there are *group straw man* arguments. In a group straw man, the arguer typically attributes some claim or belief to an entire group. For example, "Scientists are so obnoxious; they think they have all the answers!" While scientists may or may not be obnoxious, the claim about what scientists as a group think is unacceptable. The proper response is that few, if any, scientists think they have all the answers. (After all, they're in the business of searching for better answers and aren't often fully satisfied with the ones they've got.)

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False Dilemma

A *false dilemma* occurs when an arguer contends that there's a choice between two—and only two!—alternatives, one of which is clearly false or undesirable. This allows the arguer to make a convincing argument that the remaining option should be accepted. Most of the time, however, there will actually be more alternatives than the two the arguer has presented. Now, the prefix *di* in *dilemma* means *two*. But the same mistake might be made by presenting just three or four options when there are in fact more. The error isn't in the precise number of options offered, but in the failure to recognize that there may be other options that haven't yet been mentioned. The attraction of this form of argument is, again, that it requires little mental effort. The arguer doesn't need to search out creative alternative solutions, and the audience isn't asked to closely evaluate the merits of more than one decent option.

For example, in a debate, one party might turn to an onlooker and say, "Look, you're either with me or you're against me!" The unstated implication is that there's something bad about being against the arguer. The onlooker may then feel compelled to agree with the speaker since they don't want to be against them. But if we think critically, we'll realize there are definitely more than two options here. Few situations truly require that we take sides: in most debates we also have the option of being neutral or undecided.

A false dilemma might arise, for example, with regard to human resource questions, such as how to deal with an underperforming employee. Imagine you work in HR and the CEO comes to you about an employee, Martin, who's had three consecutive poor performance evaluations. The CEO says, "Either we fire Martin or we send the signal that poor performance is just fine around here!" But the situation isn't so black and white. It might be the case that the CEO hasn't spent enough time working with Martin, coaching him and setting out clear performance expectations. Or Martin might lack training, in which case it might make more sense to offer him some than to fire him. In other words, the range of options—the range of possible solutions—is more complex than the CEO is suggesting. Accepting the premise that there are just two options here would be a mistake.

The unacceptable premise that's characteristic of the false dilemma fallacy is a claim that there's a very limited range of options to choose from. A critical mindset requires that we always consider the possibility that there's something else we could do.

Slippery Slope

A *slippery slope* argument occurs when an arguer warns that one of the options available to you is liable to lead to a chain of events that will eventually end in some very bad outcome. Or that accepting a claim you're considering will logically force you to accept a chain of claims, the last

of which is clearly unacceptable. The problem is that, more often than not, those further steps do not necessarily follow from the first one.

To understand the metaphor here, picture yourself standing on level ground at the top of an icy hill. You're safe standing still where you are. But if you take just one step in the wrong direction, you'll find yourself on slippery ice, sliding down, down, down, unable to stop until you painfully crash at the bottom of the hill. In such a situation, you should of course be wary of taking that initial step if you want to avoid disaster. A slippery slope argument is one that suggests that someone facing a particular decision is, metaphorically speaking, standing at the top of a slippery slope.

Imagine a store manager saying, "We can't let this customer return that dented toaster. It works fine, and if we accept this return, we'll have to accept returns on every appliance with the tiniest dent or scratch. Next thing you know, we'll go out of business!" Or somebody saying, "If we think there's nothing wrong with people wasting their reproductive material by masturbating, we have no way of arguing against very early term abortion. But there's no real difference between early-term and late-term abortion. Before we know it, we're okay with killing babies." Don't worry here about whether you agree with this person's view; the point is to understand the kind of argument they're using. They're arguing against one belief or activity by saying that believing or doing that will inevitably lead to some further bad outcome.

We do of course need to think carefully about the short- and long-term consequences of our decisions. And with some decisions, one of the options, though perhaps desirable in its own right, might hold the possibility of quickly leading to further unpleasant consequences. An argument against experimenting with highly addictive street drugs, for instance, might constitute a reasonable slippery slope argument. When a substance is highly addictive, trying it even once might well lead to greater and greater levels of use, ending in eventual addiction and ruin. That might be a genuinely slippery slope, in which case we'd all do well to avoid it.

We commit the slippery slope fallacy, however, when we wrongly suggest that a particular "slope" is "slippery"—that is, that some option will unavoidably lead to something unpleasant when that's not likely to be the case. Whether a given slippery slope argument is fallacious or not depends on whether there's good reason to think the slope is in fact slippery. More often than not, we'll find that the slope has a number of clear "stopping points."

The unacceptable premise characteristic of the slippery slope fallacy is the claim that a slope is slippery—that choosing one option will unavoidably lead to further steps that lead to a bad outcome, or that accepting one claim will unavoidably lead to accepting further claims leading to a clearly unacceptable conclusion.

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Fallacies That Use Bad Logic

'Formal Fallacies'

The most classic examples of fallacies that rely on poor logic bear resemblance to the valid formal deductive argument types (modus ponens and modus tollens) that we have already covered. However, although they look very similar to modus ponens and modus tollens, the fallacies of *denying the antecedent* and *affirming the consequent* are in fact **invalid** argument types, which means there is a disruption in the flow of support from the premises to the conclusion.

Here are the structures for, and examples of, these two fallacies:

Denying the antecedent:

Structure:	**Example:**
If *P*, then *Q*.	*If a company is profitable, then it has at least one commercially successful product*.
Not *P*.	*This company isn't profitable*.
Therefore, not *Q*.	*Therefore, this company doesn't have a commercially successful product*.

Affirming the consequent:

Structure:	**Example:**
If *P*, then *Q*.	*If a company is profitable, then it has at least one commercially successful product*.
Q.	*This company has a commercially successful product*.
Therefore, *P*.	*Therefore, this company is profitable*.

First, as you may be able to tell by comparing these with their valid counterparts, the terms "*antecedent"* and "*consequent"* are technical terms for the first and second claims that make up a conditional claim. So, to deny the first variable is to "deny the antecedent" and to affirm the second variable is to "affirm the consequent". But what makes these fallacies?

Neither of these argument forms is deductively valid. If they were, it would be impossible to imagine a scenario in which the argument's premises are true and its conclusion false. But imagining such scenarios using these structures isn't just possible, it's actually pretty easy. As you can see, the two examples are very closely related and the same scenario can serve to show that the logical structure is faulty. We can imagine a company that has one commercially successful product but isn't profitable because it also has many unsuccessful ones. In that scenario—and considering the example of *denying the antecedent* above—it would be true that *if a company is profitable, then it has at least one commercially successful product* and true that *this company isn't profitable*, but still be false that *this company doesn't have a*

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commercially successful product. Those premises don't reliably lead to that conclusion, so the argument's underlying logical structure is flawed, or invalid. Conversely–now considering the example of *affirming the consequent*–it would be true *if a company is profitable, then it has at least one commercially successful product* and true that *this company has a commercially successful product*, but still false that *this company is profitable*. Again, the premises don't reliably lead to the conclusion, so the argument's underlying logical structure is invalid.

Another way of thinking about what's wrong with these fallacious arguments is that they mistake a necessary condition for a sufficient condition: you need at least one successful product to turn a profit (it's a necessary condition), but that doesn't mean that having such a product guarantees you'll turn a profit (it's not a sufficient condition).

If conditional arguments are invalid, the person making the argument needs to go back to the drawing board and see whether the argument can be rescued somehow. Perhaps, for example, the premises can be reformulated and used in another argument scheme. Until the person making the argument does that, there's no need to bother checking the acceptability of the premises—these arguments are no good.

On the other hand, if a conditional argument is valid, then evaluating the acceptability of the premises is vital. After all, even if the logic is good, the argument can't be good if it uses unacceptable premises as its starting point. A house can't stand on a weak foundation! And it's worth noting here that conditional claims (the foundation of every conditional argument) are often very hard to establish because such claims state that the relationship between antecedent and consequent (the two parts of the conditional claim) obtains with no exceptions. Take the previously mentioned claim *If I needed your help, I'd have called you*, for example. That's the kind of thing we routinely say in everyday life, but it's not literally true that my needing your help is a sufficient condition for my calling you; there are lots of exceptions: even if I needed your help, I wouldn't have called you if I lost my phone, if I fell into a coma, or if I'd been kidnapped, and so on.

So, establishing the truth or acceptability of conditional claims is difficult, but can be very important, especially when evaluating bridge principles. Recall that bridge principles are claims that combine a descriptive element with a normative one, and as such often take the form of a conditional claim: *if [descriptive claim], then [normative claim]*. In other words, many bridge principles specify the descriptive conditions under which normative claims are true. As bridge principles are themselves normative claims, they need to be established by finding common ground or a common purpose, which we're often able to do in business contexts. Look at this example:

Argument H:

Premise 1: *If our customers aren't satisfied with our products, we should be prepared to make changes to our product line*.
Premise 2: *Our customers aren't satisfied with our products*.
Conclusion: *We should be prepared to make changes to our product line*.

This argument is deductively valid (it's an example of modus ponens), so the question is whether the premises are acceptable? Premise 1 is a bridge principle: it tells us the descriptive conditions (*customers aren't satisfied*) under which we should accept a normative conclusion (*we should be prepared to make changes*). As with all normative claims, there's no sure-fire way to establish premise 1. But if this argument were being made at a strategy meeting, it would be easy to imagine everybody in the room accepting it, because it seems like the kind of claim that's dictated by the purpose of running a viable business.

'Informal' Fallacies

The following five inductive or informal fallacies are reasoning styles that are problematic because the logic involved is presented as strong but is revealed as weak upon closer scrutiny. In other words, what they all have in common is that the logical link between premises and conclusion is much weaker than it at first appears.

Ad Hominem (Personal Attack)

An *ad hominem* (Latin for *at the person*) or *personal attack* is the argumentative error of inappropriately focusing on the person putting an idea forward rather than on the quality of the idea itself. Doing so is often an error because, in most cases, the identity or character of the person putting an idea forward typically isn't *relevant*: a good idea is a good idea, and a bad argument is a bad argument, no matter who puts it forward. A persuasive personality may be able to make an argument *seem* stronger, but that doesn't necessarily *make* it stronger. However, as we've seen in preceding chapters, arguments can be hard to evaluate, so it's often easier to form a general opinion of a person than it is to fairly evaluate their ideas. Thus, our tendency to substitute difficult questions with easy ones can make us prone to making or accepting arguments involving this fallacy.

Imagine a colleague, Andy, suggests a change in your sales unit's accounting procedure. In response, your friend Tobi says to you, privately, "Don't listen to Andy. What does he know anyway? He's a busy-body and he didn't even donate when we were collecting for the United Way last month." Clearly, the kind of person Andy is or whether or not he donated to the United Way isn't *relevant* to the question at hand. What matters here is whether his suggestion is a good one, whether it's likely to make your unit's accounting more reliable, more consistent, and so on. Has Andy backed his advice up with good reasons? Does his advice conform to common sense? Would it be good accounting practice? The fact that this idea came from Andy is entirely irrelevant. Tobi has committed the ad hominem fallacy.

Ad hominem arguments are often more subtle than the example above. They often trade on hypocrisy or other forms of inconsistency between the attacked person's idea or argument and their behaviour. Suppose, for example, that your boss, Shauna, says she will have a piece of

software installed on everyone's computer that blocks social media during office hours, because she thinks too much work time is spent on Facebook, Twitter, and the like. Now, suppose one of your colleagues says to you "I think this is a terrible idea. I checked Shauna's Twitter and Instagram, and guess what? She posts during work hours all the time!" This is another example of an ad hominem. The error isn't as glaringly obvious as in the first example, because there's some connection between Shauna's idea and what's said about Shauna. But the fact that there's an inconsistency between the way Shauna acts and the plan she's proposing in no way implies her plan is a bad one. It just implies that Shauna might be a bit of a hypocrite—but that's a different matter.

The problem with ad hominem arguments is that what's being said (true or not) about the person putting the idea or argument forward most often provides little, if any, logical support for the conclusion that their idea or argument is a bad one. There are, however, some important exceptions. Since the main problem with ad hominem arguments regards the relevance of the personal attack to the conclusion or situation, there are some contexts where ad hominem arguments are not fallacious. Consider, for example, a political debate between potential Prime Ministers of Canada wherein one accuses another of being overly timid – "he's far too shy to be the leader this country needs!" Or someone with a criminal charge for assault applying to work in a setting where care is required – "She's far too violent to be working with those in need". In cases like this, the character of the person is relevant and important to the conclusion of the argument and thus may constitute a legitimate usage of the ad hominem.

In every case we want to ask ourselves, "Is this attack on the person relevant to the conclusion being presented?"

Appeal to Popularity

When used fallaciously, an *appeal to popularity* is the argumentative error of assuming that the mere fact that an idea is believed by many people provides decisive reason to think it's true. The problem is that the popularity of an idea has seldom has bearing on whether it's in fact true, or whether it should be used as a basis to justify our acceptance of the conclusion. An easy way to understand this is to consider historical examples. In medieval Europe, everybody "knew" the earth was flat and infectious diseases were caused by evil spirits. Yet, despite huge majorities believing them, these claims were false. Over the centuries, a great many things that were once widely believed have turned out to be false. Why is appeal to popularity so tempting? Here again, we can see substitution at work: it may be hard work to figure out whether a claim is true—much easier to simply ask whether most people believe it to be true.

In a way, an appeal to popularity is the opposite of an ad hominem. In both fallacies, the fundamental error is in turning away from the merits of the idea itself and focusing instead on some aspect of those advancing the idea (*They're a bad person, so they're wrong* or *They're the majority, so they're right*). But we should note that an appeal to popularity, while usually fallacious, often has some merit in particular situations. The fact that many people believe an idea doesn't prove that it's true, but it does provide some logical support. (You can think of it in terms of inference to the best explanation: one possible explanation for the fact that so many people believe a claim is that this claim is actually true). So, if somebody were to modestly suggest that the popularity of an idea is one consideration in its favour, this would be entirely fair. The problem arises when people overestimate the level of logical support that can be provided in this way.

What role can such mistaken reasoning play in derailing good business decision-making? Imagine a student in a co-op program questioning her workplace supervisor about the practice of calling every customer either *ma'am* or *sir*. Perhaps the student feels awkward doing so. But the supervisor replies, "Everyone in customer service knows that saying *ma'am* or *sir* is the right way to show we respect our customers." But while many may believe this, not every customer actually enjoys such formality—some people think you're calling them old when you call them *ma'am* or *sir*!

Another example demonstrating the significance of appeal to popularity can be seen when we consider successful entrepreneurs. Being critical of appeals to popularity is, in a way, at the very heart of entrepreneurship. Successful entrepreneurs are often individuals who figure out that everybody else so far has been wrong about something. If the world wrongly believes that people want *X* and you figure out that what they really want is *Y*, you may stand to make a lot of money.

The problem with appeal to popularity is that arguers tend to vastly overestimate the level of logical support that the premise *many people believe X* can lend to the conclusion *X is true*.

Again, however, there are some legitimate appeals to popularity. Can you think of any?

Appeal to Tradition

An *appeal to tradition* is the fallacy of suggesting that some belief, practice, or policy is good because it conforms to current or past practices or beliefs. To easily understand that this type of argument isn't a good one, consider that if it were a good argument, there would never be any reason to innovate. Why do something new when we've already got something that's working? But we know that innovation is often a powerful force for good. Additionally, much of what we said about appeal to popularity applies here too. The mere fact that a claim has been believed (or a practice has been followed) for a long time doesn't mean the claim is true (or the practice good). Much less does it mean that the claim should be believed (or the practice followed) going

forward—after all, circumstances may have changed. Substitution explains why appeal to tradition is tempting: it's much easier to know how things are currently done than it is to figure out the best way to do them.

But an appeal to tradition, again like an appeal to popularity, is typically not entirely devoid of logical force—the fact that a practice has been followed for a long time does provide some reason to think the practice isn't all bad (otherwise, why wasn't it abandoned?). It's just that arguers often make the mistake of thinking that an appeal to tradition is itself logically sufficient to establish a conclusion.

As a business student, it's quite likely your life will be made difficult by appeals to tradition in the near future. As you start working, armed with up-to-date knowledge about management theory and best practices, you'll likely find that your ideas about how best to run things don't always line up with the way things have been done by your new employer. Of course, you might not be in a position to do anything about that right away. But you need to remind yourself that the mere fact something has been going on for a long time isn't sufficient reason to keep doing it.

The problem with appeal to tradition is that arguers tend to vastly overestimate the level of logical support that the premise *X has been accepted for a long time* can lend to the conclusion *X is true*.

Hasty Generalization

We already discussed the fallacy of *hasty generalization* when discussing the generalizations as an argument scheme. It is the fallacy of concluding that a general statement about a population (of persons or things) is true, based on observing an insufficiently large sample.

It's important to note that we make hasty generalizations all the time; we couldn't get on with our lives otherwise. As a child, you may have learned from experience that a flame is hot. You very quickly generalized that all flames are hot, based on a very small sample size (hopefully). The problem occurs when we generalize too quickly about a complex issue. Our ability to generalize can lead us astray when we assume that a very limited amount of evidence is sufficient to allow us to reach a conclusion about a large and highly variable category.

Imagine hearing a friend say, "I bought the warranty on the last TV I bought, and that TV died after thirteen months. I was glad I had that warranty! Extended warranties on electronics are a good idea." Or a colleague say about a job applicant, "She was trained at Oxford. I worked one time with a guy who did his degree at Oxford and he was terrible! He knew lots of stuff from books, but had no common sense. We shouldn't hire her." In both cases, a strong conclusion is being reached based on very little evidence. Very few electronics fail during the extended

warranty, and not all graduates from Oxford University are the same in terms of training, skill, or character. Both of these arguers are putting forward *hasty generalizations*.

One of the most worrisome forms of hasty generalization has to do with generalizing about groups of people. Discrimination based on race, religion, sex, or sexual orientation is often based on prejudice. And prejudice (literally *pre-judging*) is often rooted in hasty generalizations about some group of people.

The problem with hasty generalization is that the arguer overestimates the level of logical support provided by a given sample size.

Post Hoc (Post Hoc Ergo Propter Hoc)

The *post hoc* fallacy occurs when an arguer suggests that because event *A* happened before event *B*, *A* must have caused *B* (*post hoc ergo propter hoc* is Latin for *after this, therefore because of this*). More generally, it's the mistake of thinking that if two things are correlated, there must be a causal relationship between them. The problem with that line of thinking is similar to the fallacious conditional arguments we discussed in chapter 10: it mistakes a necessary condition for a sufficient one. Correlation is a necessary condition for a causal relationship (if two things aren't correlated, there's likely no causal relationship between them), but it's not a sufficient one (things can be correlated even though they're not causally related). Thus, the post hoc fallacy is another instance of an argument that provides some logical support, but not as much as the arguer thinks.

The post hoc fallacy is psychologically attractive because our brains are hardwired to constantly look for causal patterns. This is generally useful, because it's often more harmful to overlook a causal relationship that actually exists than it is to see a causal relationship where there isn't one. But having this tendency comes at the expense of being prone to the post hoc fallacy, which is the root of many superstitions and conspiracy theories.

In guarding against the post hoc fallacy, it's helpful to think in terms of inference to the best explanation. After all, the reason arguments of this type offer any logical support is that *A* causing *B* does provide one explanation as to why *B* reliably follows *A*. So far, so good. But, as we know from chapter 9, it's important to consider whether there are other explanations. Consider one of the most well-known examples used in explaining the post hoc fallacy: *every morning the sun comes up shortly after my alarm goes off*. One explanation is that my alarm causes the sunrise—that's the causal explanation—but nobody thinks that it's a good explanation. (The problem is that this explanation makes for a terrible fit with relevant background knowledge.) A much better explanation is available: I know when the sun will rise, want to be awake just before then, and set my alarm accordingly. The general advice to look for alternative explanations can be made a bit

more precise in case of the post hoc fallacy. Two kinds of alternative explanation should always be considered before jumping to the conclusion that *A* caused *B*.

First, there might be a *third factor* that causes both of the phenomena. Suppose your team of consultants is asked to advise a car dealership that's experiencing declining sales. You notice that the sales started to decline shortly after most of the employees started coming to work via public transit instead of driving their own vehicles. Joan, one of your junior consultants, says, "Clearly, the employees taking transit is bad for business; perhaps people don't like to buy cars from people who don't seem to like driving themselves." Joan has committed the post hoc fallacy. She has observed that *A* was followed by *B* and immediately inferred that *A* caused *B* (and even came up with a story of how this causal relationship might work). But she hasn't considered alternative explanations. Here's one: the car dealership is in a community that has recently invested in improved public transit. This could be the cause of both *A* (employees leaving their cars at home) and *B* (the community's decreased demand for cars). This is a classic case of a third-factor explanation. The difference between these two explanations is important. If Joan's explanation is right, your team might advise the business to require its employees to drive to work. If the third-factor explanation is correct, this requirement will do nothing at all to increase demand but will do plenty to make the dealership's employees unhappy.

Second, it's important to avoid underestimating the likelihood that a correlation is simply a coincidence. Suppose you're playing a game of backgammon with your friend Sue, who always brings her own dice. When Sue rolls seven doubles in a row, you think she must be cheating—she's done something to those dice. This simply cannot be a coincidence, you think; after all, what are the chances? Well, the chances of rolling seven doubles in a row with a set of fair dice are roughly three and a half in one million. That sounds pretty low, but consider how many times someone, somewhere, rolls a pair of dice. Take a small country like Serbia, with a population of just over seven million people. If everyone in Serbia rolled a pair of dice just seven times once a year, there would be two lucky Serbs that would roll seven consecutive doubles. And their backgammon partners would likely be just as suspicious as you are of Sue. This doesn't mean we should simply accept that every weird event is a coincidence—of course not. But we need to keep the possibility in mind. Even the most unlikely events are bound to happen to somebody somewhere, so when something weird happens to us and there's no plausible causal explanation in sight, maybe it was just our turn to experience a surprising coincidence.

The problem with the post hoc fallacy is that the arguer infers a causal connection from an observed correlation without careful consideration of alternative explanations.

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Fighting Fallacies

Despite their psychological attractiveness, we're usually able to avoid falling for argumentative fallacies simply by having a critical mindset. This has something to do fallacies being errors in explicitly formulated arguments, which means we encounter them when thinking quite consciously about whether to accept claims. Having a critical mindset means approaching such situations with an inclination towards asking questions like *Is that really true? How do you know that?* and *What else are we assuming here?* Such a mindset is useful both when reading or listening to other people's arguments, and when constructing our own. Knowing about some important fallacies is helpful as, once you're aware of them, you're more likely to recognize them. And once you recognize them, you're less likely to be swayed by their psychological appeal. When listening to someone else's argument, if it gets personal, for example, you may ask, *Wait a minute—how is the person relevant here? Isn't that just a personal attack?* And when constructing your own inductive argument, a critical mindset is going to remind you to ensure you have enough information before jumping to a conclusion—in other words, it'll help you avoid making a hasty generalization.

It may also be useful to have a repertoire of standard responses to common argumentative fallacies, each of which makes explicit the problem with the argument. Table 1 provides examples. Note, however, that you need to be careful of the tone in which you challenge somebody who you believe is committing a fallacy. In most situations, it's counterproductive to challenge somebody's argument in a way that makes them feel like they're being called stupid or otherwise personally attacked.

Fallacy	**Sample Response**
Straw Man	*That's not what I said!* or *No one says that!*
False Dilemma	*You're assuming there are only two options. Let's think a bit harder about that!*
Slippery Slope	*This slope isn't really slippery. We can stop following these steps at various points.*
Ad Hominem	*The person whose idea it is isn't relevant here.*
Appeal to Popularity	*The fact that many people believe it isn't any proof. It's possible for a large number of people to be wrong.*
Appeal to Tradition	*The fact that we've always done it that way isn't a good reason. We need to look at how well we've done, whether there are better alternatives, and whether circumstances have changed.*
Hasty Generalization	*We can't make reliable claims about the population from such a small sample.*
Post Hoc	*The mere fact that there's a correlation here doesn't mean that there's a causal relationship.*

Table 1: Responding to Fallacies